IMAGE PROCESSING TECHNIQUES – EXTENDED CLASS ACTIVITIES WORKSHEET

UG Level | Duration: 10 Minutes per Activity

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Activity 4: Image Filtering (10 Minutes)

Given 5×5 Image Matrix:

[10 20 30 40 50]
[20 30 40 50 60]
[30 40 50 60 70]
[40 50 60 70 80]
[50 60 70 80 90]

Given 3×3 Mean Filter:

1/9 ×
[1 1 1]
[1 1 1]
[1 1 1]

Tasks:

1. Compute the filtered value for the center pixel (value = 50).

2. Write the formula used.

3. What type of filter is this? (Low-pass / High-pass)

4. What happens to noise after applying this filter?

Answer:

1. The 3×3 neighborhood around the center pixel is:
[30 40 50]
[40 50 60]
[50 60 70]

Filtered value = (30 + 40 + 50 + 40 + 50 + 60 + 50 + 60 + 70) / 9
= 450 / 9 = 50
Therefore, the filtered value of the center pixel is 50.

2. Formula used:
$g(x,y) = (1/9) \times \Sigma f(x+i, y+j)$, where i and j range from −1 to 1.
In simple terms, the nine pixel values in the 3×3 neighborhood are added and divided by 9.

3. Type of filter: Low-pass filter.
The mean filter is a low-pass filter because it smooths the image and reduces rapid variations in pixel intensity.

4. Effect on noise:
The mean filter reduces noise by averaging neighboring pixel values and produces a smoother image. It reduces high-frequency noise and random variations, although excessive smoothing can also blur edges and fine details.